

LAURA BURNETT, PH.D.

Research Data Scientist | Visual & Systems Neuroscience

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PROFILE

Research data scientist with ten years' experience turning large, high-dimensional biological data – imaging and video, electrophysiological time series and connectomes (graph databases) – into clear, usable results. Trained neuroscientist who designs and delivers multi-year research programmes and builds production-grade data tools in fast-paced, multidisciplinary teams. Committed to reproducibility and FAIR principles at every stage.

CORE SKILLS

Python (Pandas, NumPy, SciPy, Plotly, Dash) • SQL • Cypher (Neo4j) • HTML, CSS & JavaScript • MATLAB • Git • Experimental Design & Statistical Analysis

RESEARCH EXPERIENCE

Postdoctoral Research Associate – Investigating fruit fly visual circuits through experiments and connectomic analysis

HHMI Janelia Research Campus (Supervisor: [Michael Reiser](#)) 📅 Aug 2023 – May 2026 📍 Ashburn, VA, USA

- Designed end-to-end pipelines for high-throughput experimental screens – synchronised acquisition, cleaning and QC of large video datasets, automated documentation, statistical analysis and reproducible figure generation.
- Built and shipped an internal data product (*neuView*): a Python CLI that queries a Neo4j graph database of >150,000 cells and generates an interactive, open web resource, now deployed and used across the field alongside a landmark *Nature* (2025) study.
- Created queries and analysis for large-scale relational and graph data in SQL and Cypher.
- Worked in a multidisciplinary team: shared codebase, Git and pull-request workflow, code review, and documentation written for downstream users.

PhD Researcher – Visual attention and decision-making in mouse models of autism

Institute of Science and Technology Austria (Supervisor: [Maximilian Joesch](#)) 📅 Sep 2016 – May 2023 📍 Klosterneuburg, Austria

- Built quantitative pipelines for large, noisy multimodal data – video, behavioural and electrophysiological time series analysis and visualisation.
- Led a multi-year study on rapid decision-making in autism models, including experimental design, analysis plan, and forming a new collaboration between two research groups, building the experimental rigs and data pipelines from scratch and performing stereotactic surgeries.
- Synthesised evidence across distinct fields (*visual neuroscience*, *decision-making*, *autism*) to identify overlap and define tractable questions.

Earlier research roles – Brain and gut neural development

UCL Institute of Child Health, University of Melbourne & The Florey Institute 📅 2014 – 2016 📍 London, UK & Melbourne, Australia

Four research placements across three institutions, including quantitative analysis of confocal microscopy image data in models of enteric neuropathy and cortical development; led to publications in *Neuron*, *Nature Communications* & *Developmental Biology*.

COMMUNICATION & LEADERSHIP

- Mentoring & supervision:** mentored 11+ students (graduate, undergraduate and high-school); active member of a women & non-binary mentoring group at HHMI Janelia (since 2023).
- Speaking & stakeholders:** invited talks at international conferences and symposia (FENS Forum; Champalimaud Centre; BOKU Vienna; Janelia); elected Graduate Student Association Representative, ISTA (2018–2020).
- Teaching Assistant:** Neuroscience track course at ISTA (2018–2020); included teaching MATLAB to incoming graduate students.
- Making technical work usable:** led a coding support group and built documentation used across a lab.

OUTREACH

- STEM outreach volunteer:** active member of HHMI Janelia's RESET group (Raising Excitement for Science, Engineering and Technology) and its "Hour of Code" initiative, teaching elementary-school students to code in Python.
- Science communicator:** co-founded "Wissen Schafft's" (2021), a multilingual YouTube series explaining viruses and the immune system to the public and tackling misinformation.
- Workshop leader:** designed and led a day-long session on optical illusions for "Fakebusters", a high-school summer camp (2021 & 2022).
- Volunteer maths tutor:** Action Tutoring, Year 6 maths (since Aug 2026).

EDUCATION

MSci Natural Sciences - Neuroscience (First Class Honours)

University College London (UCL) 📅 Sep 2015 – May 2016 📍 London, UK

BSc Natural Sciences - Organic chemistry and Biomedical sciences (First Class Honours)

University College London (UCL) 📅 Sep 2012 – May 2014 📍 London, UK
University of Melbourne (study abroad) 📅 July 2014 – July 2015 📍 Melbourne, Australia

SELECTED PUBLICATIONS

h-index 5 • 397 citations (Google Scholar).

- Berg S, Beckett IR, Costa M, ... **Burnett LE**, et al. Sexual dimorphism in the complete Drosophila male central nervous system connectome. *Cell*. 2026;189(18):5504–5526.
- Nern A, Loesche F, Takemura S, **Burnett LE**, et al. Connectome-driven neural inventory of a complete visual system. *Nature*. 2025;641(8065):1225–1237.
- Burnett LE**, Koppensteiner P, Symonova O, et al. Shared behavioural impairments in visual perception and place avoidance across different autism models. *PLoS Biology*. 2024;22(6):e3002668.

FUNDING & AWARDS

Travel grant, Austrian Neuroscience Association, FENS (2022).

OeAD research scholarship: competitive, self-authored proposal (2016).

Dean's List, University of Melbourne (2014).

REFERENCES

Reference contact information is available upon request.